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Nonparametric estimation of species richness using discrete k-monotone distributions

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ABSTRACT

Nonparametric mixture models are commonly used to estimate the number of unobserved species for their robustness of modelling heterogeneity. In particular, Poisson mixtures are popular in this regard, but they are also known to have the boundary problem that causes unstable estimation. A family of shape-restricted distributions known as discrete k-monotone distributions is proposed for species richness estimation. These distributions are in fact also nonparametric mixtures and can thus be fitted rapidly via some algorithms that are made available recently. As shown by empirical studies, as compared with Poisson mixtures, their use avoids the boundary problem and gives more stable and more accurate estimates.

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1. Introduction

Estimating species richness or the number of species in a population has important implications for biodiversity management and conservation in ecology. This estimation problem is often called the "species problem". The term "species" may take on various meanings, as the problem is not unique in ecology and may occur in other fields such as biology, genetics, linguistics, software reliability; see, e.g., Chao (2005). There are many approaches to solving this problem; see Bunge and Fitzpatrick (1993) and Mao and Colwell (2005). A recent article by Barger and Bunge (2010) also provides a fully Bayesian approach, along with many references on the use of parametric *vs.* nonparametric and frequentist *vs.* Bayesian methods for this problem.

Owing to the heterogeneous nature of the problem, mixture models are commonly used to estimate the number of species. Specifically, Poisson mixtures are predominantly used (see, e.g., Böhning and Schön, 2005 and Bunge and Barger, 2008), including both the parametric case, when the mixture is finite, and the nonparametric case, when the mixing distribution is completely unspecified. For species richness estimation, especially with a nonparametric mixture, two approaches exist, which make use of the full and the conditional likelihood, respectively. Mainly for computational reasons, the conditional approach is more popular than the unconditional approach, but it suffers from an obvious drawback, i.e., the estimate of the number of species can sometimes become infinity. In comparison, the unconditional approach never gives an infinite estimate and is thus advantageous in this regard, although it may still over-estimate the number of species from time to time.

In this paper, we propose a new solution to estimating species richness, which uses discrete k-monotone distributions. By definition, a discrete k-monotone distribution is a mixture of a subfamily of discrete beta distributions. The new method is particularly suitable when a population is dominated by a large number of rare species. We focus on the unconditional

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approach to species richness estimation, for its aforementioned advantage and the availability of fast algorithms. The proposed method solves the over-estimation problem that is associated with the use of nonparametric Poisson mixtures, and, as shown in our numerical studies, is able to provide more accurate estimates.

The rest of the paper is organized as follows. We introduce the discrete k-monotone distributions and describe model parameter estimation by means of the nonparametric maximum likelihood method in Section 2. Section 3 concerns the application of mixture models to species richness estimation, in particular outlining the commonly used likelihood approaches to species richness estimation and discussing the use of the nonparametric Poisson mixture. Using discrete k-monotone models for species richness estimation is described in Section 4. A simulation study is presented in Section 5, which is followed by some real data applications in Section 6. Some concluding remarks are given in Section 7.

2. Discrete k-monotone distributions

2.1. Continuous k-monotone distributions

Prior to defining discrete k-monotone distributions, we begin with a brief introduction to continuous k-monotone distributions. The interested reader is referred to Balabdaoui and Wellner (2010) for more detail.

For an integer $k \geq 2$, a density f on $(0, \infty)$ is k-monotone if $(-1)^a f^{(a)}$, for all $a = 0, \ldots, k-2$, is nonincreasing and convex. For $k = 1$, f will simply be nonincreasing. When $k = \infty$, a k-monotone density is known to be completely monotone. An interesting characteristic of a k-monotone density is that it admits a mixture representation in which a mixing distribution enters into the specification of the density. Specifically, the integral characterization of a k-monotone density is

$$f_k(x; G) = \int_0^\infty \frac{k(\theta - x)_+^{k-1}}{\theta^k} \, \mathrm{d}G(\theta), \quad x \in (0, \infty), \tag{1}$$

where G is some distribution function on $(0, \infty)$ and $(y)_+ \equiv y\mathbf{1}_{\{y \geq 0\}}$. This mixture formulation gives a one-to-one correspondence between the family of k-monotone densities and the family of scale mixtures of beta densities with parameters 1 and k. When $k = \infty$, a k-monotone density can be represented as a scale mixture of exponential densities.

2.2. Discrete k-monotone distributions

Using the definition of a k-times monotonic sequence (see Lorch et al., 1970), one can define a discrete k-monotone distribution. Let k be a positive integer. A discrete distribution with probability mass function p on $\mathbb{N}_0 \equiv \{0, 1, \ldots\}$ is said to be k-monotone if $(-1)^a \Delta^a p \geq 0$ for all $a = 1, \ldots, k$, where Δ is the forward difference operator and the ath order difference is given by

$$\Delta^a p(x) = \sum_{i=0}^{a} \binom{a}{i} (-1)^i p(x + a - i).$$

Similarly, if $k = \infty$, a discrete k-monotone distribution is called completely monotone. An example of a discrete completely monotone distribution is the geometric distribution with success probability ϑ having probability mass function $p(x) = \vartheta(1 - \vartheta)^x$, $0 < \vartheta \leq 1$. It is known that a discrete completely monotone distribution can be represented as a mixture of geometric distributions (Steutel, 1969).

Recently, Punzo and Zini (2012) defined a discrete version of the continuous beta distribution which has a probability mass function of the form:

$$p(x) = \frac{(x + 1)^{\alpha - 1}(b + 1 - x)^{\beta - 1}}{\sum_{i=0}^{b} (i + 1)^{\alpha - 1}(b + 1 - i)^{\beta - 1}}, \quad x \in \{0, \ldots, b\},$$

where b is a nonnegative integer and $\alpha, \beta \in \mathbb{R}$. Here, we consider an extended subfamily of the discrete beta distributions given above. Specifically, let $b + 1 = \theta \in [1, \infty)$, $\alpha = 1$ and $\beta = k \in [1, \infty)$. The probability mass function of this distribution is given by

$$p_k(x; \theta) = \frac{(\theta - x)_+^{k-1}}{\sum_{i=0}^{\lfloor \theta \rfloor} (\theta - i)^{k-1}}, \quad x \in \mathbb{N}_0, \tag{2}$$

where $\lfloor \cdot \rfloor$ denotes the floor function. We shall call it the discrete monotone decreasing beta distribution. The limiting form of the discrete monotone decreasing beta distribution as $k \to \infty$ is the Dirac distribution located at the origin.

Following (1), we use the mixture form for discrete k-monotone distributions. For a given value of k, a discrete distribution is said to be k-monotone if it is a mixture of discrete monotone decreasing beta distributions (2) with probability mass

function:

$$p_k(x; G) = \int_1^\infty p_k(x; \theta)\, \mathrm{d}G(\theta) = \int_1^\infty \frac{(\theta - x)_+^{k-1}}{\sum_{i=0}^{\lfloor\theta\rfloor}(\theta - i)^{k-1}}\, \mathrm{d}G(\theta), \tag{3}$$

where G is an arbitrary mixing distribution on $[1, \infty)$. Given k, the discrete k-monotone distribution (3) is specified up to an unknown distribution function G, and since G is treated as arbitrary, (3) is in fact a nonparametric mixture distribution. When $k = 1$ (or 2), a discrete k-monotone distribution corresponds to a mixture of discrete uniform (or discrete triangular) distributions. When $k \to \infty$, the limiting probability mass function p_∞ is given by

$$p_\infty(x; H) = \lim_{k\to\infty} p_k(x; G) = \int_0^1 \vartheta(1 - \vartheta)^x\, \mathrm{d}H(\vartheta), \tag{4}$$

where H is some mixing distribution. Apparently, (4) is a mixture of geometric distributions. The limiting case (4) can be established as follows. First, using $\theta = (k - 1)\eta$, rewrite (3) as

$$p_k(x; G) = \int_{\frac{1}{k-1}}^\infty \left\{1 - \frac{x}{(k-1)\eta}\right\}_+^{k-1} \mathrm{d}\dot{G}(\eta),$$

where

$$\mathrm{d}\dot{G}(\eta) = \frac{\{(k-1)\eta\}^{k-1}\, \mathrm{d}G\{(k-1)\eta\}}{\sum_{i=0}^{\lfloor(k-1)\eta\rfloor}\{(k-1)\eta - i\}^{k-1}}.$$

Then, using $e^x = \lim_{k\to\infty}\left(1 + \frac{x}{k}\right)^k$,

$$\begin{aligned}\lim_{k\to\infty} p_k(x; G) &= \int_0^\infty e^{-\frac{x}{\eta}}\, \mathrm{d}\dot{G}(\eta)\\ &= \int_0^\infty \frac{e^{-\frac{x}{\eta}}}{\sum_{i=0}^\infty e^{-\frac{i}{\eta}}}\left(\sum_{i=0}^\infty e^{-\frac{i}{\eta}}\right) \mathrm{d}\dot{G}(\eta)\\ &= \int_0^1 \vartheta(1 - \vartheta)^x\, \mathrm{d}H(\vartheta),\end{aligned}$$

where $\mathrm{d}H(\vartheta) = \frac{1}{\vartheta}\, \mathrm{d}\dot{G}\left\{\frac{1}{\log(1-\vartheta)}\right\}$. The third equality is due to the change of variable $e^{-\frac{1}{\eta}} = 1 - \vartheta$, which leads to

$$\sum_{i=0}^\infty e^{-\frac{i}{\eta}} = \sum_{i=0}^\infty (1 - \vartheta)^i = \frac{1}{\vartheta}.$$

2.3. Maximum likelihood estimation

Suppose there is a random sample $x_1, \ldots, x_n \in \mathbb{N}_0$ from a discrete distribution, and we wish to fit by maximum likelihood a discrete k-monotone distribution to the sample. To obtain the maximum likelihood estimate of G, one needs to maximize the following log-likelihood function:

$$\ell_k(G) = \sum_{x\in\mathbb{N}_0} n_x \log\{p_k(x; G)\},$$

where $n_x = \sum_{i=1}^n \mathbf{1}_{\{x_i = x\}}$. This is in fact the problem of estimating a nonparametric mixing distribution by maximum likelihood. Many fundamental issues of the problem, such as the existence and characterization of the solution, are resolved by recognizing the likelihood maximization problem as a convex optimization problem. According to Lindsay (1983), there exists a nonparametric maximum likelihood estimate $\hat{G}$ of G that is a discrete distribution with no more support points than distinct observations in the sample. Furthermore, the nonparametric maximum likelihood estimate can be completely characterized by the gradient function

$$d_k(\theta; G) = \sum_{x\in\mathbb{N}_0} n_x \left\{\frac{p_k(x; \theta)}{p_k(x; G)} - 1\right\}.$$

An estimate $\hat{G}$ is a nonparametric maximum likelihood estimate if and only if $d_k(\theta; \hat{G}) \le 0$ for all $\theta \in [1, \infty)$. Moreover, the support points of $\hat{G}$ must lie in the set $\{\theta : d_k(\theta; \hat{G}) = 0\}$.

Although the discreteness and existence of the nonparametric maximum likelihood estimate are guaranteed, the non-identifiability issue still exists for a discrete k-monotone distribution, especially when $k \leq 2$, owing to the linearity of the component density. In particular, it means that there may exist two nonparametric maximum likelihood estimates $\hat{G}_1$ and $\hat{G}_2$, where $\hat{G}_1(\theta) \neq \hat{G}_2(\theta)$ for every θ, but $p_k(x; \hat{G}_1) = p_k(x; \hat{G}_2)$ for all x with $n_x \neq 0$. The nonuniqueness of the nonparametric maximum likelihood estimate often occurs when a perfect fit to the data is obtained, i.e., the estimated and observed probabilities are identical. When $k > 2$, the nonuniqueness problem most likely disappears, as we have never encountered it in our numerical studies. Although the nonparametric maximum likelihood estimate may be nonunique, the fitted probabilities at those distinct sample values are unique owing to the uniqueness of the mixture maximum likelihood estimate (Lindsay, 1983).

3. Species richness estimation

3.1. Two likelihood approaches

Let N denote the unknown (finite) number of species in a closed population. Suppose we obtain a single random sample of units from this population. For $i = 1, \ldots, N$, let x_i denote the abundance of species i, i.e., the number of units in the sample that belong to species i. Observe that $x_i = 0$ implies that species i has zero contribution in terms of the number of units to the sample, and hence this particular species does not present in the sample. As we only have x_i's > 0 in the sample, the distribution of the observed species abundances is truncated at zero. For $x \in \mathbb{N}_1 \equiv \{1, 2, \ldots\}$, let $n_x = \sum_{i=1}^{N} \mathbf{1}_{\{x_i = x\}}$ denote the number of species that contributes exactly x units to the sample. We call $\{n_x : x \in \mathbb{N}_1\}$ the set of observed counts. The total number of observed species in the sample is then given by $n_{\text{obs}} = \sum_{x \in \mathbb{N}_1} n_x$. Species richness estimation refers to the estimation of N based on the observed species abundances or the observed counts.

For modelling the N abundances, one can consider a mixture model. Under mixture modelling, each abundance has its own abundance parameter. For instance, the abundance parameter under Poisson mixture modelling can be interpreted as the mean abundance. It is assumed that the N abundance parameters can be described by N independent and identically distributed random variables with some common distribution. Conditional on the abundance parameter value, the abundance of a species is assumed to have some simple distribution, such as the Poisson distribution. Hence, unconditionally, the N abundances are assumed to come from a mixture distribution. If we assume that the N abundances are a random sample from a mixture distribution having probability mass function $p(x; G)$ on $\mathbb{N}_0$, then $\{n_x : x \in \mathbb{N}_0\}$ is a random sample from a multinomial distribution with probability mass function given by

$$p(n_0, n_1, \ldots; N, G) = \frac{N!}{\prod_{x \in \mathbb{N}_0} n_x!} \prod_{x \in \mathbb{N}_0} \{p(x; G)\}^{n_x} .$$

Since n_0 can be replaced with $N - n_{\text{obs}}$, the full likelihood function can be written as

$$L(N, G) = \frac{N!}{(N - n_{\text{obs}})! \prod_{x \in \mathbb{N}_1} n_x!} \{p(0; G)\}^{N - n_{\text{obs}}} \prod_{x \in \mathbb{N}_1} \{p(x; G)\}^{n_x} . \tag{5}$$

Only the nonzero observed counts have actual contributions to the full likelihood. As given in (5), there are two parameters, a parameter of interest N and a nuisance distribution function G. If G is a degenerate distribution, then all N abundances have the same abundance parameter value and the population is said to be homogeneous. Homogeneity assumption always does not hold in real applications. Thus, mixture models are highly valuable as they naturally cope with heterogeneity.

Sanathanan (1972) showed that (5) can be decomposed into two parts:

$$\begin{aligned} L(N, G) &\equiv L_m(N, G) \times L_c(G) \\ &= \frac{N!}{(N - n_{\text{obs}})!\, n_{\text{obs}}!} \{p(0; G)\}^{N - n_{\text{obs}}} \{1 - p(0; G)\}^{n_{\text{obs}}} \times \frac{n_{\text{obs}}!}{\prod_{x \in \mathbb{N}_1} n_x!} \prod_{x \in \mathbb{N}_1} \left\{ \frac{p(x; G)}{1 - p(0; G)} \right\}^{n_x} , \end{aligned}$$

where $L_m(N, G)$ is the marginal likelihood given n_{obs} and $L_c(G)$ the conditional likelihood given $\{n_x : x \in \mathbb{N}_1\}$ (conditioning on n_{obs}). The set of cell probabilities in the conditional likelihood in fact forms a zero-truncated mixture distribution, implying that conditioning on n_{obs}, the observed species abundances are a set of observations from a zero-truncated mixture distribution.

Because of the above decomposition, there are two popular likelihood approaches, conditional *vs.* unconditional, to species richness estimation. The conditional approach is a two-stage procedure, in which the best fitting model to the truncated data is first obtained, and then an extrapolation is made to predict N on the basis of the best fitting model and the observed species richness n_{obs}. Specifically, this approach first seeks the (conditional) nonparametric maximum likelihood estimate $\hat{G}_c$ from $L_c(G)$. Then, since $L_m(N, G)$ is a binomial likelihood, given $\hat{G}_c$, the integer-valued maximum likelihood estimate of N from $L_m(N, \hat{G}_c)$ is given by $\hat{N}_c = \lfloor n_{\text{obs}}/\{1 - p(0; \hat{G}_c)\} \rfloor$; see Sanathanan (1972, 1977). Recently, Böhning and

Kuhnert (2006) demonstrated that for the problem of conditional maximum likelihood estimation, one can directly work with the likelihood for a mixture of zero-truncated distributions rather than that for a zero-truncated mixture of untruncated distributions since the estimates of N using these two distributions are the same; see also Mao and Lindsay (2007).

The unconditional approach, on the other hand, searches for the (unconditional) estimates $\hat{N}_u$ and $\hat{G}_u$ that jointly maximize the full likelihood; again, see Sanathanan (1972, 1977). In addition, Wang and Lindsay (2005) noted that if $(\hat{N}_u, \hat{G}_u)$ maximizes the full likelihood, then $\hat{N}_u = \lfloor n_{\text{obs}}/\{1-p(0, \hat{G}_u)\}\rfloor$. Thus, both approaches can be considered similar in the sense that a nonparametric maximum likelihood estimate (conditional or unconditional) is first computed and then the estimate of N can be obtained from a simple equation.

Fast algorithms for nonparametric and semiparametric mixture estimation are proposed by Wang (2007) and Wang (2010b), respectively.

3.2. Using nonparametric Poisson mixture models

Species richness estimation using the nonparametric Poisson mixture model by the unconditional maximum likelihood approach was first considered by Norris and Pollock (1998). A time-consuming algorithm was used by the authors in which the unconditional estimates $\hat{N}_u$ and $\hat{G}_u$ are found by maximizing the profile likelihood function of N. To be specific, for a sequence of fixed N integer values, (5) with $p(x; G)$ being the probability mass function of the Poisson mixture distribution is maximized over G for each N value, one at a time. Then $(\hat{N}_u, \hat{G}_u)$ is the pair that maximizes the profile likelihood $L(N, \hat{G}_N)$ over N, where $\hat{G}_N$ is the nonparametric maximum likelihood estimate for a given N. Recently, Böhning and Schön (2005) proposed an algorithm for maximizing the zero-truncated Poisson mixture likelihood and estimating the number of species. As mentioned earlier, in the conditional maximum likelihood approach, one can actually consider maximizing the likelihood for the mixture of zero-truncated Poissons. By reparametrizing a zero-truncated mixture of untruncated Poisson distributions into a mixture of zero-truncated Poisson distributions, the problem of computing the nonparametric maximum likelihood estimate can be solved in a more elegant way.

Despite being a popular means of modelling count distributions, the nonparametric Poisson mixture model has some nontrivial practical difficulties in the context of species richness estimation. These include the boundary problem and instability issue (Wang and Lindsay, 2008); see also Wang and Lindsay (2005) and Kuhnert et al. (2008). The boundary problem refers to the problem that arises in N estimation due to a phenomenon associated with the nonparametric maximum likelihood estimate. In practice, it is frequently observed that the nonparametric maximum likelihood estimate tends to include the left boundary point of the parameter space into its set of support points. Consider the conditional approach to N estimation using the mixture of zero-truncated Poisson distributions in which the mixing distribution is defined on $(0, \infty)$. If the boundary point 0 is excluded, it is possible that the nonparametric maximum likelihood estimate would never be attained since this violates the compactness assumption needed to guarantee the existence of the nonparametric maximum likelihood estimate (see Lindsay, 1983). Including it, however, could cause a severe consequence for the estimate of N. If the nonparametric maximum likelihood estimate puts some probability mass at 0 or near 0, then the estimate of N reaches ∞ or is highly inflated. On the other hand, 0 is a legitimate boundary point in the unconditional approach to N estimation using the nonparametric Poisson mixture model. However, a terrible over-estimation of N could occur if the nonparametric maximum likelihood estimate happens to place a high probability mass in the neighbourhood of 0. Thus, the estimate of N based on the nonparametric maximum likelihood estimate has an instability issue. These provide the motivation for the penalized approach that penalizes the conditional likelihood, as put forward by Wang and Lindsay (2005) and Wang (2010a).

4. Species richness estimation using discrete k-monotone distributions

4.1. Using discrete k-monotone distributions

Generally speaking, the observed species abundance distribution is right-skewed, often with the highest count at $x = 1$. To provide an adequate fit to such a distribution, the mixture of zero-truncated Poissons does not always seem to be a strong candidate. Even if it is suitable for modelling the observed species abundance distribution, it can still perform badly in estimating the number of species as the nonparametric maximum likelihood estimate could possess some very small abundance parameter value, leading to an over-estimation of N. Furthermore, in the presence of some large outlying nonzero counts, the Poisson mixture assumption might not be able to offer a smooth fit in the tail of the observed species abundance distribution. Importantly, while it is convenient to assume Poisson sampling errors for count data, as with normal sampling errors for continuous data, this assumption may not be suitable in all situations.

We propose estimating the number of species using discrete k-monotone models. In studying the conditional and unconditional approaches to species richness estimation, we find that the former indeed poses some practical estimation problems. One of the problems is that, for a particular data set, the nonuniqueness of the nonparametric maximum likelihood estimate obtained from maximizing the likelihood for a mixture of zero-truncated discrete monotone decreasing beta distributions results in different estimates of N. Hence, we only describe the unconditional maximum likelihood approach to species richness estimation using discrete k-monotone models.

For a given k-value, the discrete k-monotone model assumes that the N abundances are independent and identically distributed observations from a mixture of discrete monotone decreasing beta distributions and that the N abundance parameters follow an unspecified distribution G. With k fixed, the task now is to maximize the following log-likelihood function:

$$\ell_k(N, G) = \log N! - \log (N - n_{\text{obs}})! - \sum_{x\in\mathbb{N}_1} \log n_x! + (N - n_{\text{obs}}) \log \{p_k(0; G)\} + \sum_{x\in\mathbb{N}_1} n_x \log \{p_k(x; G)\}. \tag{6}$$

Although N is naturally a positive integer, for easier likelihood maximization, we extend its range to positive real numbers, and hence evaluate a factorial involving N with the gamma function. With this slight extension, the unconditional estimation problem requires one to find $\hat{N}_u^*$ (generally not an integer) and $\hat{G}_u^*$ that maximize (6).

Since (6) is a "semiparametric" log-likelihood, we adopt the CNM-MS algorithm of Wang (2010b) for computing $(\hat{N}_u^*, \hat{G}_u^*)$. It is an efficient algorithm for finding the maximum likelihood estimate of a semiparametric mixture model. If one really wants an integer solution, then one can simply use the nearest integer to $\hat{N}_u^*$, or compute the profile likelihood $L(N, \hat{G}_N)$ at $N = \lfloor \hat{N}_u^* \rfloor$ and $\lceil \hat{N}_u^* \rceil$ (where $\lceil \cdot \rceil$ denotes the ceiling function), respectively, and choose the one giving the higher profile likelihood value.

4.2. Model selection

In considering the usage of the discrete k-monotone model for estimating the number of species, one might be concerned that the nonparametric maximum likelihood estimate is not always unique. However, when the nuisance nonparametric maximum likelihood estimate is not unique, the integer solution of N is still unique provided that $\lfloor \hat{N}_u^* \rfloor$ and $\lceil \hat{N}_u^* \rceil$ do not have the same maximized profile likelihood value. Accordingly, we argue that the nonuniqueness of the nuisance nonparametric maximum likelihood estimate does not pose any significant alarm as far as the estimation of the number of species is concerned.

In practice, an appropriate value of k for a particular data set is unknown. As the family of discrete k-monotone distributions become more shrunken when k increases, $\ell_k(\hat{N}_u^*, \hat{G}_u^*)$ decreases monotonically with k. Moreover, increasing k typically tends to reduce the number of support points. Owing to these conditions, we need a selection tool to control the balance between the goodness of fit and the model complexity. Our pragmatic approach is to repeatedly fix a k-value while estimating G and then select k based on the Akaike information criterion. We note that Böhning and Schön (2005) considered the evaluations of candidate mixture models by the Akaike and the Bayesian information criterion in the context of species richness estimation. Wang and Chee (2012) found that while both the Akaike and the Bayesian information criterion appear to perform reasonably well for choosing an appropriate value for bandwidth in the context of mixture-based density estimation, the Akaike information criterion seems more reliable. Selecting k is a model selection problem and can also be guided by a cross-validation criterion, for example.

4.3. Dealing with high abundances

A pattern that is often noticeable in the truncated species abundance distribution is the presence of relatively large amounts of rare species but small amounts of abundant species. According to Barger and Bunge (2010), all existing estimation methods are affected by sparse nonzero counts in the right tail. Typically, a right truncation point τ splitting the observed distribution into rare species and abundant species groups is chosen. Then, the abundant species group or the observed counts beyond τ are initially ignored in the estimation, i.e., the initial estimation of N is solely based on $\{n_x : x = 1, \ldots, \tau\}$. Finally, the total number of abundant species is added to that estimate of N. See also Wang and Lindsay (2005) and Bunge and Barger (2008).

Since completely ignoring the abundant species group may cause severe bias to the resulting estimate of N, we offer a proposal for handling the truncated species abundance distribution with high abundances by slightly adjusting the observed counts in the full likelihood. In particular, for some large τ, we add up all the observed counts for $x > \tau$ and use their total $n_{\tau+} = \sum_{x=\tau+1}^{\infty} n_x$. The full likelihood function thus becomes

$$L_\tau(N, G) = \frac{N!}{\prod_{x=0}^{\tau+} n_x!} \prod_{x=0}^{\tau+} \{p(x; G)\}^{n_x}, \tag{7}$$

where $p(\tau+; G) = 1 - \sum_{x=0}^{\tau} p(x; G)$.

4.4. Overcoming instability

As the largest count is often observed at $x = 1$ in the truncated species abundance distribution, one would expect the probability at 0 to be high. A Poisson mixture often has to include a mean parameter value near 0 in order to obtain a large probability at 0, and hence can easily result in an over-estimated N.

Numerical evidence given below shows that as compared with the estimates of N based on nonparametric Poisson mixture models, those based on discrete k-monotone models do not suffer from the instability problem, because of their

Table 1
Component distributions for the simulation study.

Distribution	Notation	Probability mass function	Parameters
Poisson	$\text{Pois}(\theta)$	$\frac{\theta^x \exp(-\theta)}{x!}$	$\theta > 0$
Negative binomial	$\text{Nbin}(\gamma, \theta)$	$\frac{\Gamma(\gamma+x)}{x!\Gamma(\gamma)} \left(\frac{\gamma}{\theta+\gamma}\right)^{\gamma} \left(1 - \frac{\gamma}{\theta+\gamma}\right)^{x}$	$\gamma > 0,\ \theta > 0$
Geometric	$\text{Geom}(\theta)$	$\theta(1-\theta)^x$	$\theta \in (0, 1]$

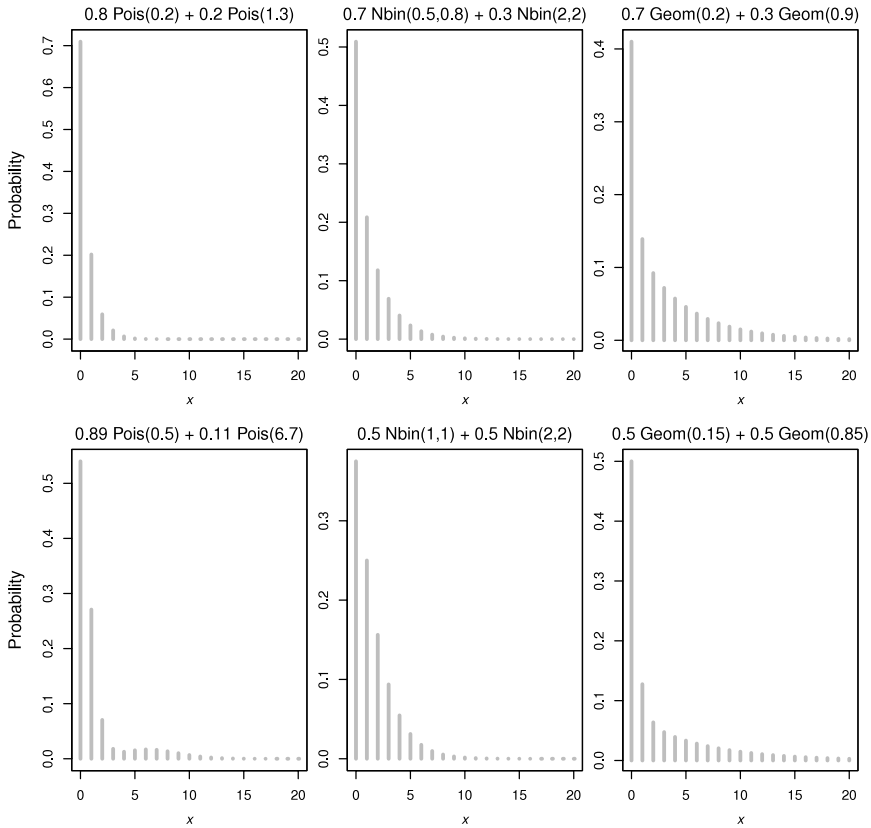

Fig. 1. Six different data generation settings considered in the simulation study.

enforced shape restrictions. As a result, the nonparametric maximum likelihood estimate of a discrete k-monotone model tends not to include support points close to the left boundary point, which is the main cause of over-estimating N. Specifying a larger k-value typically pushes the support points further away from the left boundary point but does not necessarily produce a higher fitted probability at 0. Therefore, the estimate of N does not necessarily increase with k.

5. A simulation study

Five approaches to species richness estimation are compared in our simulation study. Denoted by UL–PM, PCL–PM, PCL–PCG, LS–TM and UL–k, they are, respectively, the unconditional likelihood approach using the Poisson mixture model (Norris and Pollock, 1998), the penalized conditional likelihood approach using the Poisson mixture model (Wang and Lindsay, 2005, 2008), the penalized conditional likelihood approach using the Poisson-compound gamma model (Wang, 2010a), the least squares approach using the triangular mixture model (Durot et al., 2014) and our unconditional likelihood approach using the discrete k-monotone model.

Three types of component distributions, i.e., Poisson, negative binomial and geometric, as listed in Table 1, are used in our simulation study, and 200 data sets were generated from each of the six two-component mixture distributions shown in Fig. 1. The two Poisson mixtures were previously used in the simulation study of Wang (2010a). All these six settings should likely favour the nonparametric Poisson mixture and the Poisson-compound gamma, rather than our discrete k-monotone distributions, since the six mixtures used for data generation all belong to the family of Poisson mixture distributions. In every setting, the observed counts are obtained from after removing all the zeros in $N = 1000$ generated data points.

Table 2
Grid points used for Akaike information criterion evaluation.

2.01	2.5	3	3.5	4	4.5	5	5.5	6	6.5
7	7.5	8	8.5	9	9.5	10	–	–	–
11	12	13	14	15	16	17	18	19	20
21	23	25	27	29	31	33	35	37	39
41	45	49	53	57	61	65	69	73	77
81	89	97	105	113	121	129	137	145	153
161	177	193	209	225	241	257	273	289	305
321	353	385	417	449	481	513	545	577	609
641	705	769	833	897	961	1025	1089	1153	∞

To select a value for the tuning parameter k of the discrete k-monotone distribution, we employ the Akaike information criterion. The values tabulated in Table 2 are potentially considered as the grid points for Akaike information criterion evaluation. To speed up the selection process, we always first find the fit with $k = \infty$ and then the fits with an increasing k, starting at $k = 2.01$. The fitting process stops if a fit has the same number of parameters as the one with $k = \infty$, as no fit in between can have a smaller AIC value than it. The Poisson-compound gamma model also has a tuning parameter, and the grid points used here are the same as in the simulation study of Wang (2010a), i.e., 1–10, 20, 40, 60, 100 and ∞. Moreover, the specification of a value for τ is required by PCL–PM and PCL–PCG, and we use the default settings in Wang (2011).

The simulation results are summarized in Table 3. For comparison, we calculate the same performance measures as Wang and Lindsay (2005), i.e., the sample mean ($\bar{N}$), the median ($\widetilde{N}$), the standard deviation (s), the root mean squared error (RMSE), the mean absolute error (MAE) and the central 95% percentile interval (95%). As far as the unconditional likelihood approach is concerned, the discrete k-monotone model performs very well in every case and very much dominates the Poisson mixture model, in terms of the closer averaged estimates $\bar{N}$ and $\widetilde{N}$ to the true N, the lower variability, the smaller error scores and the narrower central 95% interval. Comparing our estimator with the other three estimators, we can see that it has less bias but suffers from high variance, and hence only occasionally outperform the others in terms of RMSE and MAE. On the other hand, its central 95% interval, as well as that of the Poisson-compound gamma estimator, captures the true N in every setting, whereas the other two do not. Despite our estimator fares as well as the Poisson-compound gamma estimator, it enjoys a very significant computational advantage over this competitor, as shown in the next section.

6. Real applications

Four real data sets are further studied, two with n_0 known (but removed for estimation) and two with n_0 unknown. The results are given in Table 4. For each data set, we provide two plots in Fig. 2, showing the discrete k-monotone fit to the data, along with the estimates of n_0 by the other four approaches, and the gradient function evaluated at the nonparametric maximum likelihood estimate. Each fit by the k-monotone distribution is nearly perfect. The intersection points between the gradient curve and the x-axis, marked by solid dots in the bottom panels, are the support points of the nonparametric maximum likelihood estimate.

Specifically, the accident data contain a set of counts from an insurance company. For $N = 9461$ issued accident insurance policies, let n_x denote the number of policies reporting exactly x claims during a particular year. The nonzero counts are $n_0 = 7840$, $n_1 = 1317$, $n_2 = 239$, $n_3 = 42$, $n_4 = 14$, $n_5 = 4$, $n_6 = 4$ and $n_7 = 1$. The data set has been previously studied by Böhning and Schön (2005) giving 5503 as an estimate of N, which is apparently a severe under-estimation. As compared with the other approaches, the estimate of N using the discrete 9-monotone model, although it still seemingly under-estimates N, is much closer to the observed value.

The filarial worms data set is the other real example with N known exactly. The data contain the frequencies of filarial worms on rat mites (Bertram, 1949). With the count at zero omitted, the observed species abundance distribution has the highest count of 553 at $x = 1$. There are also a number of species with high abundances found in the data set. In other words, the number of filarial worms found on some mites is large. The observed species richness n_{obs} is 1445 and the species richness N is 2600, implying that the number of mites with no filarial worms is 1155. Parametric models were demonstrated by Heller (1997) for being able to estimate N excellently. The discrete k-monotone model, with $k = 19$ selected, gives an estimate of 2787 for N, much closer to the observed value than those obtained by the other approaches.

The microbial data set, previously used by Bunge and Barger (2008), is concerned with the microbial diversity and the interest is in the number of microbial species. The microbial data contain 149 observed species richness and the highest count at $x = 1$ is 67, with the true N unknown. The observed species abundance distribution exhibits the presence of some sparse observations in the right tail. Bunge and Barger (2008) reported an estimate of 308 for N. For this data, the selected discrete 27-monotone model and the approach of Wang (2010a) give estimates of N that match closely with that by Bunge and Barger (2008). On the other hand, the other three approaches coincidentally yield the same estimate of N, namely 254.

The last data set concerns vocabulary richness estimation, in which the number of unknown species is the number of words that Shakespeare knew, as studied previously by Efron and Thisted (1976). Since we notice that the data summarized in Table 1 in Efron and Thisted (1976) do not tally with the counts in Appendix A of Spevack's (1968) concordance, we recalculate the numbers of words up to 100 occurrences directly from Spevack (1968) and give them in Table 5. The actual counts for the occurrences beyond 100 are also available, with a total of 849 words. Our counting shows that Shakespeare

Table 3
Comparison of simulation results for five approaches to species richness estimation. The true value of N is 1000 for all settings.

Approach	$\bar{N}$	$\widetilde{N}$	s	RMSE	MAE	95%
	0.8 Pois(0.2) + 0.2 Pois(1.3)					
UL–PM	3453.65	1033.56	6641.86	7064.99	2716.46	(551.34, 25756.18)
PCL–PM	784.99	779.00	138.13	255.37	227.38	(552.90, 1081.25)
PCL–PCG	929.93	935.00	174.29	187.44	142.62	(574.85, 1262.33)
LS–TM	637.09	637.00	37.85	364.87	362.91	(566.75, 714.00)
UL–k	938.00	864.66	413.89	417.48	244.79	(602.77, 1791.73)
	0.89 Pois(0.5) + 0.11 Pois(6.7)					
UL–PM	1124.69	1051.48	301.78	325.82	161.67	(897.24, 2085.14)
PCL–PM	1066.08	1049.00	118.99	135.85	101.31	(898.80, 1336.90)
PCL–PCG	1046.69	1032.00	111.85	120.94	88.73	(889.80, 1275.62)
LS–TM	935.75	939.00	38.81	75.01	66.04	(854.98, 1017.00)
UL–k	1000.10	978.06	93.68	93.44	65.38	(884.67, 1136.53)
	0.7 Nbin(0.5, 0.8) + 0.3 Nbin(2, 2)					
UL–PM	1000.39	754.99	1467.41	1463.74	418.80	(629.19, 1788.11)
PCL–PM	771.42	761.00	89.00	245.21	229.46	(635.00, 959.25)
PCL–PCG	844.34	849.50	102.03	185.99	164.94	(653.70, 1054.47)
LS–TM	791.45	787.50	38.38	212.03	208.55	(722.98, 865.08)
UL–k	896.16	853.37	159.30	189.82	155.22	(758.29, 1258.65)
	0.5 Nbin(1, 1) + 0.5 Nbin(2, 2)					
UL–PM	1124.96	889.61	1396.46	1398.56	316.51	(789.43, 2250.65)
PCL–PM	928.93	906.50	99.47	122.05	104.89	(798.90, 1182.03)
PCL–PCG	1008.08	1029.00	121.63	121.59	99.48	(815.98, 1229.28)
LS–TM	973.04	972.00	39.70	47.90	39.61	(901.92, 1056.03)
UL–k	1058.56	1035.62	154.56	164.92	70.82	(944.28, 1344.45)
	0.7 Geom(0.2) + 0.3 Geom(0.9)					
UL–PM	841.57	745.61	793.25	806.96	303.17	(659.59, 1130.95)
PCL–PM	764.64	755.00	65.11	244.16	235.78	(667.88, 910.20)
PCL–PCG	806.79	794.50	81.12	209.47	195.97	(685.90, 1002.60)
LS–TM	775.57	774.50	30.03	226.42	224.44	(713.98, 839.02)
UL–k	877.21	832.48	169.51	208.97	181.14	(723.99, 1374.17)
	0.5 Geom(0.15) + 0.5 Geom(0.85)					
UL–PM	757.06	706.68	294.06	380.87	300.61	(596.80, 1087.47)
PCL–PM	707.80	703.00	69.41	300.29	293.55	(605.98, 848.48)
PCL–PCG	830.30	798.50	161.92	234.28	207.31	(613.83, 1312.25)
LS–TM	688.04	687.50	28.74	313.27	311.96	(628.92, 750.00)
UL–k	990.67	832.78	631.13	629.62	296.73	(678.82, 2567.86)

Table 4
Species richness estimation results for four real data sets.

Approach	Estimate of N			
	Accident ($N = 9461$)	Filarial worms ($N = 2600$)	Microbial	Shakespeare
UL–PM	5496	2214	254	72 586
PCL–PM	5496	2213	254	72 195
PCL–PCG	6935	2844	304	73 004
LS–TM	4016	2286	254	56 050
UL–k	7374	2787	296	112 605

Table 5
Shakespeare word count data. Tabled value n_x is the number of distinct words occurring exactly x times. 14 376 distinct words occur just once each, 4343 twice each, 2292 thrice each, and so on.

x	1	2	3	4	5	6	7	8	9	10
0+	14 376	4343	2292	1573	1043	835	638	518	430	363
10+	305	259	243	223	187	181	178	130	127	128
20+	104	105	99	112	93	74	83	76	72	63
30+	73	48	56	59	53	45	33	49	44	51
40+	49	41	30	35	37	21	41	30	28	19
50+	25	19	28	27	31	19	19	22	23	14
60+	29	19	21	18	15	10	15	14	11	16
70+	13	12	10	16	18	11	8	15	12	8
80+	13	12	11	8	10	11	7	12	9	8
90+	4	7	6	7	10	10	15	7	7	5

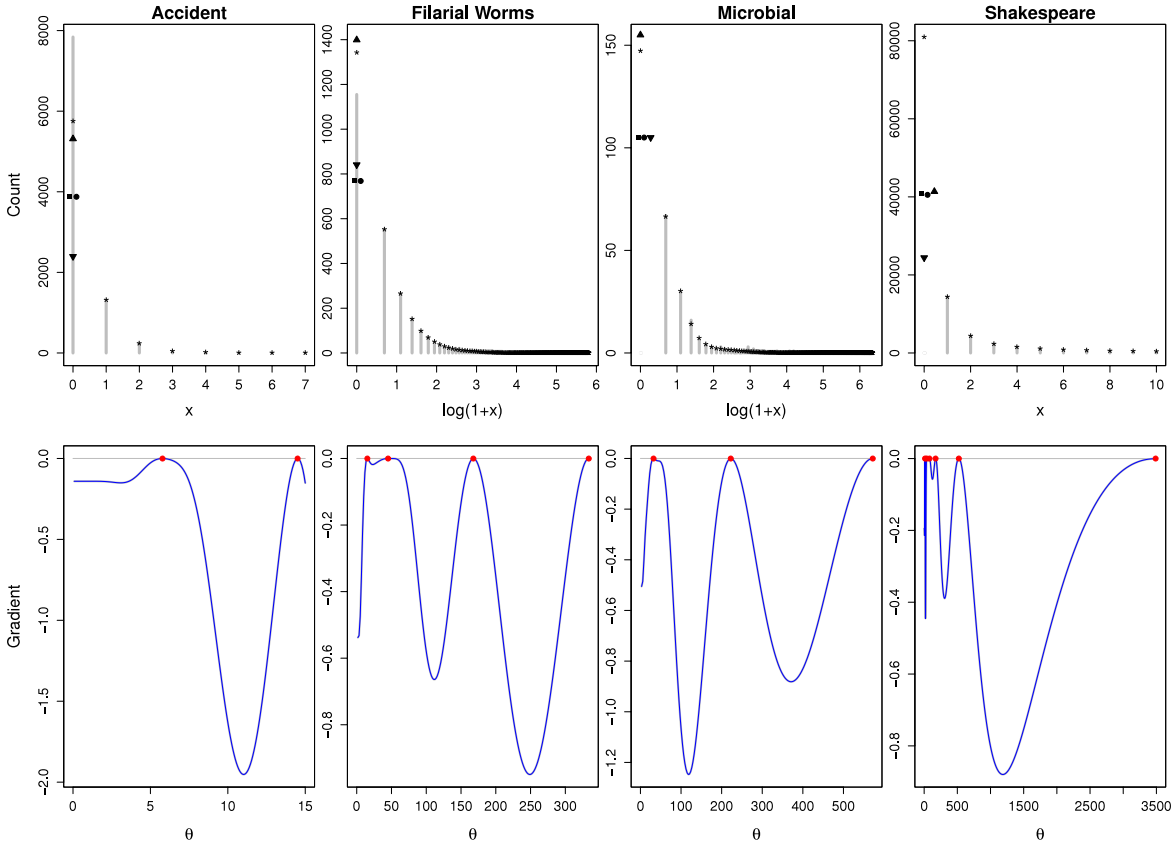

Fig. 2. The top panels show the observed (vertical bar) *vs.* estimated (⋆) counts based on the fitted discrete k-monotone model, along with the estimates of n_0 by UL–PM (■), PCL–PM (●), PCL–PCG (▲) and LS–TM (▼). The bottom panels show the gradient curve evaluated at the nonparametric maximum likelihood estimate for each data set.

used a total of 884 650 words in his works. There are a large number of rare words and quite a number of extremely abundant ones. Specifically, the observed vocabulary richness is 31 641 of which 14 376 occur just once each, 4343 twice each, 2292 thrice each, and so on. For this real example, in order to estimate N using the Poisson mixture and discrete k-monotone models, we consider maximizing the full likelihood (7) and set $\tau = 100$. Similarly, we use $\tau = 100$ for LS–TM. On the other hand, we use $\tau = 50$ for PCL–PM and PCL–PCG since the largest effective τ-value in Wang's (2011) implementation is 50. The result using the 25-monotone model indicates that the richness of Shakespeare's vocabulary is 112 605 words, about twice the estimate of N by the approach of Durot et al. (2014). Moreover, our approach suggests that Shakespeare knew about 40 000 more words than estimated by the other three approaches.

For the first two of the four real data sets, our estimates are the closest to the known n_0's, and for the third one, it is similar to that of Wang (2010a). For the fourth one, a referee questioned the accuracy of our estimate, since our estimate ($112\,605 = 80\,964 + 31\,641$) is much greater than the others. Except the estimate by LS–TM which is apparently an under-estimate, the other three are similar and all reasonably close to the lower bound ($35\,000 + 31\,641$) provided by Efron and Thisted (1976). We argue that this lower bound is too low and that our estimate is very reasonable, based on two simple extrapolation analyses as follows. First, we apply a log transformation to n_x and calculate $\log n_x - \log n_{x+1}$ for $x \geq 1$ (see the left panels of Fig. 3). Apparently, $\log n_x$ has a general trend that is not only decreasing with x but also convex. Therefore, a linear extrapolation based on $\log n_1$ and $\log n_2$ should give an under-estimate of $\log n_0$, or n_0. This under-estimate of n_0 is 47 586, as shown correspondingly in circle in the top left panel of Fig. 3, which gives 79 227 for N, an estimate already greater than those found by the other methods, including the lower bound by Efron and Thisted (1976). The second extrapolation analysis is similar, except that we use a log–log–log–log transformation, as shown in the right panels of Fig. 3. The general trend of $\log(\log(\log(\log n_x)))$ is also clearly both decreasing and convex, and hence the linear extrapolation from the log–log–log–log-values at $x = 1$ and 2 should still give an under-estimate of n_0. This second under-estimate of N is 102 271, which is far greater than the other estimates but still smaller than ours (112 605). Our approach is essentially also an extrapolation that deals with the monotone trend in a more refined way.

To end this section, we report in Table 6 the computing times of all approaches for obtaining the results in Table 4, using a computer with a 2.66 GHz Intel Core 2 Quad processor and 4 GB of RAM. Clearly, our approach is also computationally competitive.

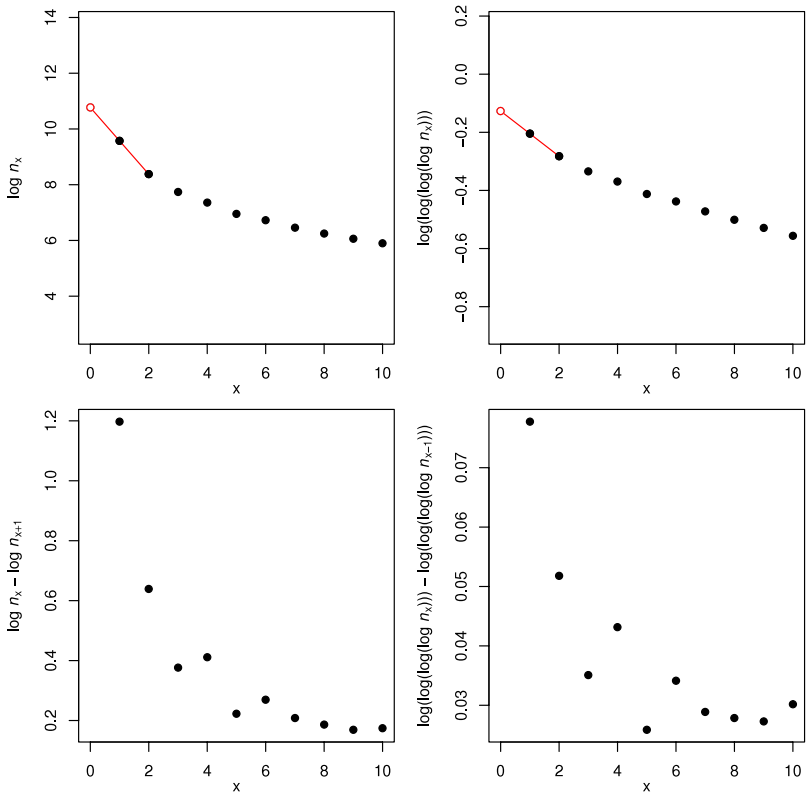

Fig. 3. Top panels: Plots of log transformation and log–log–log–log transformation of observed counts (solid dots) and an estimated count (circle). Bottom panels: The corresponding plots of differences between two consecutive points.

Table 6
Computing times (in seconds) of all five approaches for four real data sets.

Approach	Time			
	Accident	Filarial worms	Microbial	Shakespeare
UL–PM	0.55	2.91	0.86	17.60
PCL–PM	0.17	234.63	3.17	1925.11
PCL–PCG	33.43	6259.88	3558.80	58485.46
LS–TM	0.01	3.04	2.73	5.95
UL–k	5.46	41.92	21.74	225.64

7. Summary and remarks

In this paper, we introduce discrete k-monotone distributions and use them for estimating species richness. These distributions are in fact nonparametric mixtures, with discrete monotone beta distributions being mixture components. Since the number of unobserved species is certainly unknown and to be estimated, the estimation here turns out to be the same as that of a semiparametric mixture, which has efficient algorithms available for likelihood maximization. We leave the value of k to be determined by a model selection criterion and found in particular that the Akaike information criterion seems to work well. The use of discrete k-monotone distributions does not suffer from the boundary problem, as by nonparametric Poisson mixtures, and is hence able to provide more stable and more accurate estimates of the number of species.

There have been relatively few studies among research articles dealing with species richness estimation that have considered the unconditional likelihood approach. Here, we take an attempt to maximize the full likelihood function of both the nonparametric mixing distribution function and the total number of species. It differs from the conditional likelihood approach, where the maximized likelihood function is conditional on the number of observed species. It is also computationally inexpensive, in aid of the efficient algorithms available now for fitting semiparametric mixture models. As evidenced in the numerical studies, our approach has a huge significant gain in computational efficiency over the approach of Wang (2010a), which is really a strong and close competitor. Last but not least, unlike the other approaches, our approach is very suitable for observed species abundance distributions with extreme skewness.

Using discrete k-monotone distributions enforces certain shape restrictions on the distribution function to be estimated. While this is good for its nonparametric nature, such restrictions cannot be appropriate in all situations and the resulting estimators may turn out to be biased, possibly significantly in some cases. In principle, however, it is possible to test the goodness of a fitted model and, if the fit is deemed inappropriate, one can gradually decrease the level of distributional restrictions and refit the model until a good fit is obtained. How to turn such general considerations into exact solutions is an interesting topic for future research.

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References

Balabdaoui, F., Wellner, J.A., 2010. Estimation of a k-monotone density: characterizations, consistency and minimax lower bounds. Stat. Neerl. 64, 45–70.
Barger, K.J.-A., Bunge, J.A., 2010. Objective Bayesian estimation for the number of species. Bayesian Anal. 5, 765–786.
Bertram, D.S., 1949. Studies on the transmission of cotton rat filariasis. I. The variability of the intensities of infection in the individuals of the vector, *Liponyssus bacoti*, its causation and its bearing on the problem of quantitative transmission. Ann. Trop. Med. Parasitol. 43, 313–332.
Böhning, D., Kuhnert, R., 2006. Equivalence of truncated count mixture distributions and mixtures of truncated count distributions. Biometrics 62, 1207–1215.
Böhning, D., Schön, D., 2005. Nonparametric maximum likelihood estimation of population size based on the counting distribution. J. R. Stat. Soc. Ser. C. Appl. Stat. 54, 721–737.
Bunge, J.A., Barger, K.J.-A., 2008. Parametric models for estimating the number of classes. Biom. J. 50, 971–982.
Bunge, J.A., Fitzpatrick, M., 1993. Estimating the number of species: a review. J. Amer. Statist. Assoc. 88, 364–373.
Chao, A., 2005. Species estimation and applications. In: Balakrishnan, N., Read, C.B., Vidakovic, B.D. (Eds.), Encyclopedia of Statistical Sciences. Vol. 12, second ed. Wiley, New York, pp. 7907–7916.
Durot, C., Huet, S., Koladjo, F., Robin, S., 2014. Nonparametric species richness estimation under convexity constraint. arXiv:1404.4830.
Efron, B., Thisted, R.A., 1976. Estimating the number of unseen species: how many words did Shakespeare know? Biometrika 63, 435–447.
Heller, G.Z., 1997. Estimation of the number of classes. South African Statist. J. 31, 65–90.
Kuhnert, R., Del Rio Vilas, V.J., Gallagher, J., Böhning, D., 2008. A bagging-based correction for the mixture model estimator of population size. Biom. J. 50, 993–1005.
Lindsay, B.G., 1983. The geometry of mixture likelihoods, part I: a general theory. Ann. Statist. 11, 86–94.
Lorch, L., Muldoon, M.E., Szego, P., 1970. Higher monotonicity properties of certain Sturm–Liouville functions. III. Canad. J. Math. 22, 1238–1265.
Mao, C.X., Colwell, R.K., 2005. Estimation of species richness: mixture models, the role of rare species, and inferential challenges. Ecology 86, 1143–1153.
Mao, C.X., Lindsay, B.G., 2007. Estimating the number of classes. Ann. Statist. 35, 917–930.
Norris III, J.L., Pollock, K.H., 1998. Non-parametric MLE for Poisson species abundance models allowing for heterogeneity between species. Environ. Ecol. Stat. 5, 391–402.
Punzo, A., Zini, A., 2012. Discrete approximations of continuous and mixed measures on a compact interval. Statist. Papers 53, 563–575.
Sanathanan, L., 1972. Estimating the size of a multinomial population. Ann. Math. Statist. 43, 142–152.
Sanathanan, L., 1977. Estimating the size of a truncated sample. J. Amer. Statist. Assoc. 72, 669–672.
Spevack, M., 1968. A Complete and Systematic Concordance to the Works of Shakespeare, Vols. 1–6. George Olms, Hildesheim.
Steutel, F.W., 1969. Note on completely monotone densities. Ann. Math. Statist. 40, 1130–1131.
Wang, J.-P.Z., 2010a. Estimating species richness by a Poisson-compound gamma model. Biometrika 97, 727–740.
Wang, J.-P.Z., 2011. SPECIES: an R package for species richness estimation. J. Stat. Softw. 40, 1–15.
Wang, J.-P.Z., Lindsay, B.G., 2005. A penalized nonparametric maximum likelihood approach to species richness estimation. J. Amer. Statist. Assoc. 100, 942–959.
Wang, J.-P.Z., Lindsay, B.G., 2008. An exponential partial prior for improving nonparametric maximum likelihood estimation in mixture models. Stat. Methodol. 5, 30–45.
Wang, Y., 2007. On fast computation of the non-parametric maximum likelihood estimate of a mixing distribution. J. R. Stat. Soc. Ser. B Stat. Methodol. 69, 185–198.
Wang, Y., 2010b. Maximum likelihood computation for fitting semiparametric mixture models. Stat. Comput. 20, 75–86.
Wang, Y., Chee, C.-S., 2012. Density estimation using non-parametric and semi-parametric mixtures. Stat. Model. 12, 67–92.